



Simple Triple Sine Wave Generator

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This generator produces audible signals to test microphones and codecs used in audio amplifier equipment. This simple triple sine wave generator can also be used for testing other audio systems, including VoIP (Voice over Internet Protocol) equipment.

The equipment for transmitting speech signals is usually designed to transmit audio signals in the range of around 300Hz to 3400Hz. Testing of such equipment is usually carried out using three frequencies—300Hz, 1000Hz, and 3400Hz. During the

tests, we normally use separate and mixed frequencies.

Circuit and working

Circuit diagram of the simple triple sine wave generator is shown in Fig. 1. It is built around three dual audio operational amplifiers NE5532 (IC1 through IC3), a dual power supply, and a few other components.

You can use an operational amplifier like NE5532A, RC4560, or any similar or better one capable of driving

The Frequencies of the Simple Triple Sine Wave Generator

Resistors	Capacitors	Frequency (approximate)
R4 = R5 = 5.1k	C1 = C2 = 100nF	312Hz
R6 = R7 = 4.7k	C3 = C4 = 33nF	1026Hz
R8 = R9 = 4.7k	C5 = C6 = 10nF	3400Hz

PARTS LIST

Semiconductors:

IC1-IC3 - NE5532 operational amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R3 - 68-ohm
R4, R5 - 5.1-kilo-ohm
R6-R9, R14, R19 - 4.7-kilo-ohm
R10-R13, R16, R18 - 10-kilo-ohm
R15 - 30-kilo-ohm
R17, R20-R23 - 100-ohm
VR1-VR3 - 510-ohm potmeter
VR4-VR6 - 1-kilo-ohm potmeter

Capacitors:

C1, C2, C7, C8, C11-C14 - 100nF ceramic disk
C3, C4 - 33nF ceramic disk
C5, C6 - 10nF ceramic disk
C9-C10 - 100 μ F, 40V electrolytic

Miscellaneous:

CON1-CON2 - 2-pin connector
CON3 - 3-pin connector
- 8-pin base (3 number)
- Jumper wires

(All these are available from kitsnspares.com)

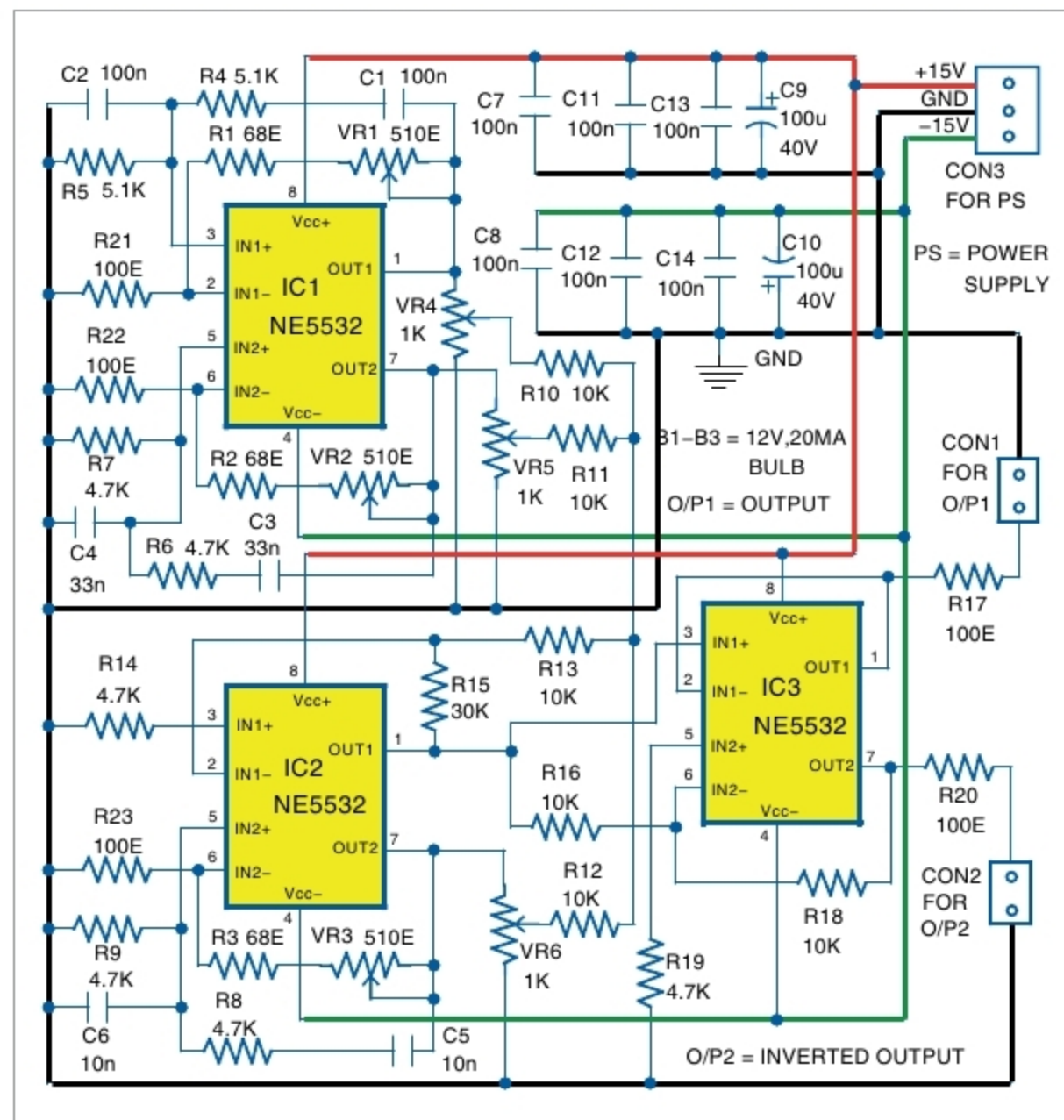


Fig. 1: Circuit diagram of triple sine wave generator

loads of up to 600-ohm.

The circuit has three sine wave generators with Wien bridges and three operational amplifiers (IC1 through IC3). The frequencies of the oscillators are shown in the table above.

It is better to use resistors R4 through R9 and capacitors C1 through C6 with tolerance of ± 2 per cent or better. We can calculate the frequency of oscillations by using the formula for the standard Wien bridge oscillator.

Pins 1 and 7 of IC1 are outputs of